MEDICATION SAFETY SIGNAL MONITOR · HAND-IN R02

Pipelines: raw to ontology

Version 1.1, 5 November 2026: how four public sources become a ranked, traceable review queue

The question this report answers

How does raw public data become the ranked signals, recalls and changes the safety officer sees, and how would another engineer change it?

One-line answer

Four public sources land as raw copies; drug identity is resolved once; FAERS, labels and recalls are cleaned by written rules; curated datasets back the Ontology; a statistics step and a 3-step label check (AI terms, exact text, reviewed synonyms) produce 1,073 ranked signals to review. Every rule lives in one file, so a change has a known ripple.

What this is	What this is not
• The pipeline end to end, in the order data flows • The rules (decisions D0xx) and why each one exists, with real examples: oxycodone and semaglutide • A guide for an engineer who must add a drug, add a source or change a rule	• A Foundry primer (see R01 for datasets, transforms, Ontology and Workshop) • A list of every data problem (see R07, data judgment) • A medical finding: a signal is disproportionate reporting plus *not found in the label*, not proof that the drug causes the reaction • Reviewed by a pharmacist (pending)

How to read this. Chapter 1 is the whole pipeline on one page. Chapters 2 to 7 follow the data from source to curated datasets. Chapter 8 is the signal statistics. Chapters 9 and 10 cover what runs incrementally, the change feed and monitoring. Chapter 11 is for the next engineer. Each chapter ends with *how we know*. The evidence labels are printed at the end of chapter 1. Codes (D### decision, A### approach, DQ## data issue, O# optimization) and *step N* (a step of the 12-step build order; step 9 = first full 13-drug build) are explained in R01 chapter 3. Four wide diagrams are printed sideways on their own page so they can be read.

Contents	Page
01 · The pipeline in one page	3
02 · Four sources and the raw contract	6
03 · Drug identity, resolved once	6
04 · Cleaning FAERS	7
05 · Labels and the AI reader	8
06 · Recalls	9
07 · Curated datasets: what backs the Ontology	10
08 · Signal statistics and the label check	10
09 · What runs incrementally, and what it saved	14
10 · Change feed and monitoring	14
11 · How another engineer extends this	18
A · Where the numbers come from	20

01

The pipeline in one page

The safety officer watches 13 drugs: 8 opioids and 5 GLP-1 drugs (diabetes and weight loss). Opioids: oxycodone, hydrocodone, morphine, fentanyl, hydromorphone, methadone, buprenorphine, tramadol. GLP-1 drugs: semaglutide, liraglutide, dulaglutide, tirzepatide, exenatide. She asks three questions each morning: which drugs have side effects reported that the label does not mention; which have an active recall; and what changed since yesterday. Every flag must trace back to its source.

Stage	What happens	Main output	Size
1 · Raw	Pull 4 public sources exactly as received, with request, status and time	`raw_*`	1,321 labels, 900 recalls
2 · Identity	Resolve drug names to one ingredient key, once, for every later step	`clean_drug_identity`	13 ingredients
3 · Clean	FAERS versions, deletes, roles, duplicates; label safety sections; recalls and NDC matches	`clean_faers_*`, `clean_label_*`, `clean_recall*`	792,346 current cases
4 · Curated	One row per object; link tables; AI label reading (pilot)	`ontology_*`	10 object types
5 · Signals	2×2 counts, PRR, Evans rule, 3-step label check, buckets, rank	`ontology_signal_signals`	1,952 signals, 1,073 ranked
6 · Change + health	Dated change events; source as-of; 21 checks per build	`ontology_change_events`, `monitoring_*`	2,898 events

Three ideas hold it together DESIGNED:

- **Resolve drug identity once, in clean.** Every count, label and recall joins on the same ingredient key (RxNorm RxCUI), so no step re-guesses a drug name.
- **Write each rule down and give it a version.** The signal rule is `prr_evans_v2`; the label check is `text+ai_v1+syn_v2`. A decision the officer makes is tied to the exact computation she saw (an `evaluation_id`).
- **Keep what you cannot use.** Unmatched names, error replies, mass reports and uncertain label matches are kept with a reason, never dropped silently.

How to read the headline numbers

A **signal** (1,952) is a drug–reaction pair that meets the Evans rule (at least 3 reports, PRR at least 2, chi-squared at least 4; D006, proposed) **and** whose reaction is not found in any reference label of that drug (1,844 *not in label*) or is found only through a medium-band synonym (108 *possibly in label*).

Ranked (1,073) is the review queue: signals that are *not in label*, in the clinical bucket (D016 v3) and less than half carried by mass-polypharmacy reports (D023 v2). It is a review queue, not 1,073 findings, and it does not show that a drug causes a reaction. It was 1,355 at step 9 (the first full build); chapter 11 shows how it fell.

What this means for Dana (the safety officer)

Signals move only with a new FAERS quarter. FAERS arrives as quarterly files; at the baseline gate the newest data was about 130 days old (monitoring check `faers_age_days`). So *what changed since yesterday* almost never contains a new signal; it is mostly recalls and label versions. FAERS moved to the FDA's new AEMS system in March 2026; the future feed is not verified (DQ20).

The Morning brief change card counts the last 30 days (26 events), not since yesterday. Page 3's date range narrows it to *since my last review*.

The AI does not touch her top signal. The AI read only 2 oxycodone pilot labels; for semaglutide × NAION it read none, so *not in label* there comes from the exact-text and synonym checks (chapter 8). NAION is a *known positive*, not a discovery: EMA PRAC (6 Jun 2025) and a WHO alert (27 Jun 2025) named it, and our 3 US labels do not mention it (DECISIONS learning 2026-11-05). Only the project owner can submit a triage decision today.

Gate: Chapter 1: the pipeline answers the three questions TESTED

Required evidence: Each question is fed by a named dataset with a count from `numbers.json`.

Result: Met: Q1 1,073 ranked signals, Q2 126 active recalls, Q3 26 changes in the last 30 days (baseline gate, 5 Nov 2026) OBSERVED.

If it fails: The baseline gate (`docs/11-handin/baseline-gate.md`) is re-run before any hand-in.

Label	Meaning
OBSERVED	Read directly from a public source (FAERS, DailyMed, openFDA, RxNorm), with the query that produced it.
DERIVED	Computed by our pipeline from observed data, for example a PRR or a count per drug and reaction.
ASSUMED	Not verified: taken as true for now because the data or the brief is silent. It could be wrong.
DESIGNED	A design choice we made and wrote down (the decision log says why). It may be built; the status band says how far.
TESTED	Built, run, and checked against evidence.
PLANNED	On the plan; not built yet.
PROPOSED	Suggested; waiting for a decision.

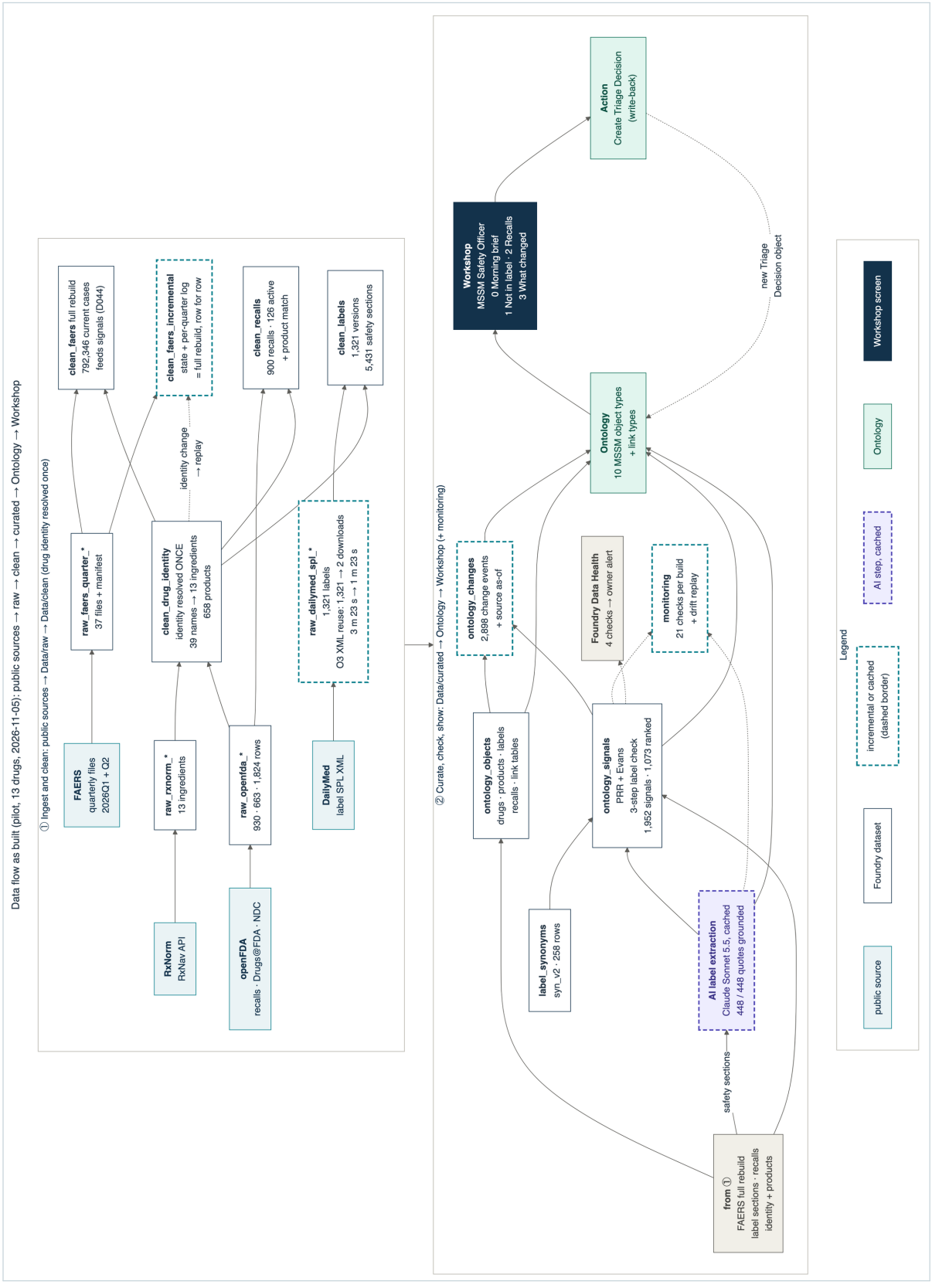


Figure 1. Data flow as built (v2, O3 measured): public sources → `Data/raw` → `Data/clean` (identity resolved once) → `Data/curated` → Ontology → Workshop. Dashed borders are incremental or cached steps. The product count in the identity box comes from the README, not from `numbers.json`. (Printed sideways on its own page so the labels can be read.)

Source: docs/diagrams/png/30-handin-data-flow-v2.png (source docs/diagrams/src/30-handin-data-flow-v2.mmd)

02

Four sources and the raw contract

Figure 1 (the sideways page before this chapter) shows the whole flow from the four sources to the screen.

Source	What it is	Raw datasets	Rows OBSERVED
FAERS quarterly files	FDA Adverse Event Reporting System: spontaneous reports of suspected side effects. Bulk text files, 2026 Q1 and Q2	`raw_faers_quarter_files`, `_manifest`	files + manifest
DailyMed (NLM)	Official drug labels as SPL XML (Structured Product Labeling), with versions	`raw_dailymed_spl_index`, `_xml`	1,321 labels
openFDA (FDA API)	Recalls (enforcement reports); Drugs@FDA applications; NDC package directory	`raw_openfda_enforcement`, `_drugsfda`, `_ndc`	930 · 663 · 1,824
RxNorm (RxNav API)	US standard drug vocabulary: names, ingredients, salts, related concepts	`raw_rxnorm_lookup`, `_related`	13 ingredients

The raw contract

- Keep the original reply (JSON, XML or file) beside a few parsed columns.
- Save request parameters, HTTP status, retrieval time and the source's own *last updated*.
- A non-200 reply is an **error row**, never a silent gap.
- Scope is one `INGREDIENTS` list at the top of each pull file (13 names since step 9).
- After every build: row count = the source's own total (`meta.results.total`, file bytes), per drug.

How we know

All raw counts equal the source totals for each of the 13 drugs, with 0 error rows (A013).

The count check caught two bugs that raise no error: an openFDA OR search without parentheses returned 11 oxycodone recalls instead of 25, and searching only the harmonised `openfda.*` field found 47 Drugs@FDA applications instead of 141 (DECISIONS learnings, 2026-11-04).

03

Drug identity, resolved once

FAERS names drugs in free text: brands, misspellings, doses inside the name. Labels and recalls belong to *products*, not ingredients. So identity is built once, in `clean_drug_identity.py`, and every later step joins on its outputs DESIGNED (D020; D027, proposed).

Output	Key	Rule
`clean_drug_ingredients`	`ingredient_rxcui`	The D018 formulary: 13 ingredient RxCUIs (an explicit list, because ATC classes miss methadone and hydrocodone, DQ31)
`clean_drug_identity`	`lookup_string`	D020: exact RxNorm ingredient name, then strip salt words (HYDROCHLORIDE, CITRATE …). **Never substring.** Unmatched names kept

Output	Key	Rule
`clean_drug_product_map`	`ingredient_rxcui` + `product_key`	Product = FDA application number (D039 G2, a check inside decision D039); combination products flagged
`clean_drug_package_map`	`package_ndc11`	11-digit NDC (National Drug Code); no application → `labeler:` key
`clean_label_product_map`	`setid` + `spl_version` + `application_number`	Approval ids read from the label XML; one *reference label* per product = the application owner's label (G3)

Why never substring OBSERVED: matching *morphine* as a substring also catches apomorphine and diamorphine (heroin). In the October prototype (local Python scripts that applied the same rules to the same FAERS files on 20 October, before we had Foundry) that made morphine rows about 50% too high (DQ34).

The two levels (D027, proposed): counts belong to the ingredient (Drug); labels and recalls belong to the product (application) and package (NDC). Bulk FAERS carries an application number (`nda_num`), so a report can reach the right label (D024): for example, injected semaglutide and the oral tablet have different labels (DQ33).

```
Oxycodone chain (trace T13):
RxCUI 7804 → application NDA022272 (OXYCONTIN) → package 59011-410-10
          → reference label bfdfe235…, version 44
```

How we know

Each identity dataset has 0 duplicate keys (Foundry SQL). The OxyContin chain above returns exactly one row. The local check `foundry/tools/local_run/check_identity.py` repeats both tests.

04

Cleaning FAERS

`clean_faers.py` reads both quarters and writes three outputs OBSERVED: `clean_faers_cases_current` (792,346 cases), `clean_faers_case_reactions` (2,554,017 rows, every current case, needed as the background for PRR) and `clean_faers_drug_reaction` (373,522 rows for our 13 drugs). Four rules, in order:

Rule	What it does	Why (evidence)
D022 versions and deletes	Per quarter: keep the highest `caseversion` per `caseid` (newer quarter wins ties), then apply that quarter's delete list to *all* history	FAERS sends follow-ups in later quarters. Q2 replaced 26,681 Q1 cases and deleted 655. Appending would double count (DQ36) OBSERVED
D021 suspect roles	Count a drug when it is primary suspect (PS), secondary suspect (SS) or interacting (I); once per case; never when only concomitant	For oxycodone, primary suspect only would miss 56% of cases; SS often repeats PS (DQ35)
D020 identity	Split `prod_ai` on `\`, resolve each part through `clean_drug_identity`; blank → whole `drugname`	One identity rule for every source (chapter 3)
D023 v2 duplicates and mass reports	Fold the same person filed by many companies; then leave every report listing **30 or more drugs** out of the main count, but keep it as `a_mass_reports`	A Canadian mass-polypharmacy cluster: 465 of 3,408 oxycodone cases, median 122 drugs, ages changing between filings. It put pemphigus at the top of the list (A012)

Example: oxycodone OBSERVED. 4,464 suspect cases before the duplicate fold, 3,408 after (foundry/README). The ≥30-drug cut then sets 465 of those aside as mass reports.

How we know

Counts equal the October prototype exactly (792,346 cases; oxycodone 4,464 → 3,408; DECISIONS learnings, 2026-11-04). The prototype is our own earlier code, so agreement shows the Foundry port is faithful. It does not show the rules are right; the rules are argued in D020 to D023.

The incremental path (chapter 9) applies the same rules quarter by quarter and gives identical rows, which tests D022 independently.

05

Labels and the AI reader

DailyMed is the label source of truth (D019): openFDA's label copy lags it (DQ28). `clean_labels.py` writes `clean_label_versions` (one row per setid + version, effective date from the XML, modern PLR format flag) and `clean_label_sections`.

Safety section	LOINC code	Note
Boxed Warning	34066-1	
Contraindications	34070-3	
Warnings and Precautions	43685-7	Modern (PLR) labels
Warnings / Precautions	34071-1 / 42232-9	Older labels use two separate sections (DQ19)
Adverse Reactions	34084-4	Sub-sections 6.1/6.2 use a generic code, so everything nested is folded in (DQ40)

DQ42: a 200 that is not a label OBSERVED. For two tramadol repackager labels DailyMed answered HTTP 200 with the text `None`. The parser crashed. Now raw keeps the reply as received; clean keeps only text that starts with `<`; the version stays listed with `has_xml = false`.

The AI reader (pilot)

`ai_label_extraction.py` sends each safety section to Claude Sonnet 5.5 through AIP and asks for every reaction with an **exact quote**. Code checks the quote is in the section text. Results: 2 labels, 9 sections, 448 reactions, 448 grounded OBSERVED. The cache key is setid + version + section + text hash + prompt, parser and model version, so an unchanged label costs 0 model calls (O1).

Watch out

The AI is a pilot on 2 oxycodone labels and does not change the signal list today: the text check already covers its terms. It read no semaglutide label, and its quote is not on the officer's screen yet. Widening it waits for a pharmacist-checked answer key (D041).

How we know

Grounding: 448 of 448 quotes found word for word, across all 9 sections OBSERVED.

Accuracy is a smaller test: F1 0.97 (proposed D015 target 0.85) on a 35-item answer key, the Boxed Warning and Contraindications of the 2 oxycodone labels, written by Claude and frozen before the run, no pharmacist yet. The model found 35 of 35 answer-key items (`eval/ai-eval-v1.md`).

Label versions: 1,321, 0 version mismatches against the DailyMed index.

06

Recalls

`clean_recalls.py` writes `clean_recalls_current` (900 recalls; the latest pull wins; status must be Ongoing, Completed or Terminated) and `clean_recall_product_match`. A recall is **active** when its status is Ongoing: 126 today OBSERVED.

NDCs come from the recall's own text DESIGNED. openFDA attaches every package of an application to every recall record, so its `openfda.package_ndc` field over-matches. We parse NDCs from the product description and code info instead, and grade each match:

Confidence	Meaning	Used for the Recall ↔ Product link?
high	A full package NDC printed in the recall, found in the NDC directory	yes
medium	A product NDC, or a 10-digit NDC resolved to one package	yes
low	Only the ingredient name appears in the text	no (feeds Recall ↔ Drug instead)
none	No NDC in text, NDC not in the directory, or an ambiguous 10-digit NDC; kept with a reason	no

Example: the newest recall, D-0040-2027 (reported 2026-10-21), is a compounded tirzepatide product. It names our drug, so it shows under tirzepatide, but it matches none of our products, and the screen says so (*Matched to our products: 0*). A 2027 number with a 2026 date is not a typo: we take the year to be the FDA's fiscal year, which starts on 1 October ASSUMED.

How we know

Every recall has an ingredient-level row, so none disappears for lack of an NDC. In the oxycodone pilot, 20 of 31 printed package NDCs were no longer in the NDC directory, but all 4 active recalls matched a listed package (DECISIONS learnings, 2026-11-04).

07

Curated datasets: what backs the Ontology

`ontology_objects.py` turns clean data into one row per object and one row per link, in `Data/curated`. Keys are checked for nulls and duplicates before Foundry indexes them OBSERVED.

Curated dataset	Object or link	Key
`ontology_drug_drugs`	MSSM Drug (13)	`ingredient_rxcui`
`ontology_product_products`	MSSM Product, with its reference label	`product_key`
`ontology_label_labels`	MSSM Label (1,321), with DailyMed URL	`label_key` = setid|v<version>
`ontology_recall_recalls`	MSSM Recall (900)	`recall_number`
`ontology_link_*` (4 tables)	Drug↔Product, Product↔Label, Recall↔Product, Recall↔Drug	`link_key`
`ontology_labeled_reactions`	MSSM Labeled Reaction (448, AI)	`labeled_reaction_key`
`ontology_ae_stats`	MSSM AE Stat (14,921)	`stat_key`
`ontology_signal_signals`	MSSM Signal (1,952)	`signal_key` = rxcui|reaction|rule
`ontology_change_events`, `ontology_source_status`	MSSM Change Event, MSSM Source Status	`event_id`, `source`
`ontology_triage_decisions`	MSSM Triage Decision: empty; rows come from the Action	`decision_id`

The curated layer is where *grain* is fixed: one signal per drug × reaction × rule; one label per setid × version. The Ontology adds nothing the dataset does not hold, except the officer's decisions DESIGNED (D040).

> **How we know**
> Object counts in Foundry equal the dataset rows; 0 null and 0 duplicate keys in all curated tables (A013).

08

Signal statistics and the label check

Figure 2 (after step D) follows one FAERS report to a ranked signal in four stages: clean the reports (A), count and test (B), check the label (C), sort the queue (D). It is printed in two parts so the boxes can be read.

Step B: count and test

For each drug × reaction, `ontology_signals.py` builds a 2×2 table: *a* = reports with this drug and this reaction; *b* = this drug, other reactions; *c* = other drugs, this reaction; *d* = the rest. **PRR** (proportional reporting ratio) = (a / all reports for the drug) ÷ (c / all reports for other drugs), with a 95% confidence interval and a Yates chi-squared. The **Evans rule** flags a ≥ 3, PRR ≥ 2 and chi-squared ≥ 4 (D006, proposed). Output: 14,921 drug × reaction rows (*stats*) in `ontology_ae_stats`; diagram 32 counts 2,533 of them meeting Evans.

Step C: is it in the label? Three checks

Every stat is checked against the safety sections of the drug's **reference labels** (one per product, the application owner's label; D039 G3). The code tries the checks in the order below and stops at the first hit (`ontology_signals.py` l.163–176). The outcome would be the same in any order; only the recorded `label_match_method` differs. The version string `text+ai_v1+syn_v2` names the rule versions, not the order.

Check	Rule	Outcome
1 · AI terms (D041)	An atomic term the AI extracted, with a quote the code found word for word, equals the FAERS term. Only the 2 oxycodone pilot labels have AI terms	in label
2 · Text (D037, proposed)	The exact FAERS term or its US spelling appears in a safety section of the drug's reference labels	in label
3 · Synonyms (D043)	A reviewed phrase for the term appears in those sections. Band **high** → in label; **medium** → *possibly in label* (kept, not ranked); **rejected** rows kept for audit	in / possibly

The synonym table `label_synonyms.py` (version syn_v2) has 258 rows: 69 high, 72 medium, 117 rejected. A script proposes leads; a person sets the band (so far Claude; pharmacist pending).

Step D: from 1,952 signals to 1,073 ranked

Group	Count	Rule (`ontology_signals.py` l.194–200)
Signals	1,952	Evans **and** (not in label **or** possibly in label)
· not in label	1,844	No check found the term in any reference label
· possibly in label	108	Found only through a medium-band synonym: kept with a caution, never ranked
Ranked (the review queue)	1,073	Not in label **and** clinical bucket **and** mass-report share under 0.5
Not in label but not ranked	771 DERIVED	In a tagged bucket (D016 v3: abuse/overdose, product use and errors, efficacy, pregnancy, social/legal, administrative) or mostly from mass reports (diagram 32 counts 231 signals with that caution)

Buckets are keyword groups (D016 v3), a stand-in for MedDRA groupings we cannot license. The **mass-report share** is a_mass ÷ (a + a_mass), where a_mass counts reports listing 30 or more drugs, which were left out of *a* (D023 v2, `ontology_signals.py` l.196). A share of 0.5 or more means the reaction is mostly carried by those re-filed reports, so it gets a caution instead of a rank.

Ranked signals are ordered by chi-squared within each drug (D042). D042 records the choice, not a comparison: chi-squared grows with both the size and the strength of the excess, while ordering by the PRR's lower confidence bound is an alternative we have not tested DESIGNED. Every signal carries an `evaluation_id`, a hash of its key, count, PRR, label check, window, number of labels searched and number of AI-read labels, so a decision is pinned to what the officer saw DESIGNED.

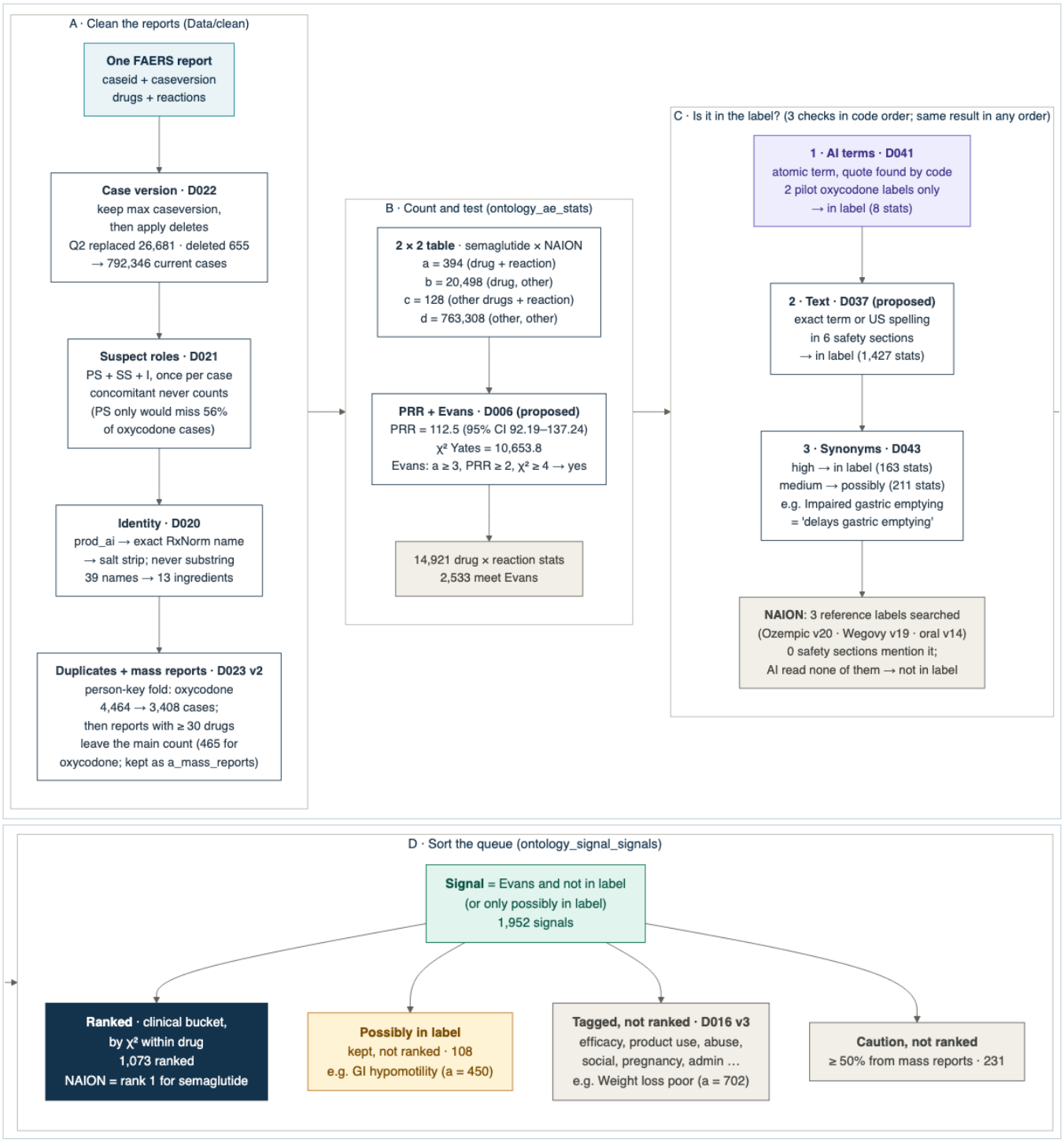


Figure 2. From one FAERS report to a ranked signal (v2), with real numbers for semaglutide × NAION. Top: stages A to C. Bottom: stage D. Read two units with care. **Stats** in stage C are all 14,921 drug × reaction rows, flagged or not (for example *possibly (211 stats)*). **Signals** in stage D are only the Evans rows that are not, or only possibly, in the label (for example *possibly in label · 108*). The NAION box counts the 3 *reference labels* searched; the text below also quotes a wider manual look at all 36 safety sections of every semaglutide label in DailyMed (A014).

Source: docs/diagrams/png/32-handin-signal-pipeline-v2.png (source docs/diagrams/src/32-handin-signal-pipeline-v2.mmd)

Worked example: semaglutide × NAION

Item	Value	Evidence
Reaction (MedDRA preferred term)	Optic ischaemic neuropathy (NAION)	OBSERVED
a: reports with semaglutide and NAION	394 of 20,892 semaglutide reports	OBSERVED
PRR (95% CI)	112.5 (92.19 – 137.24)	DERIVED
Chi-squared (Yates)	10,653.8	DERIVED
Label check	Not found in the 3 reference labels searched (Ozempic v20, Wegovy v19, oral Ozempic v14). The AI read none of them, so this comes from the text and synonym checks. A wider manual look found NAION in none of the 36 safety sections of any semaglutide label in DailyMed (A014)	OBSERVED
Rank for semaglutide	1	DERIVED

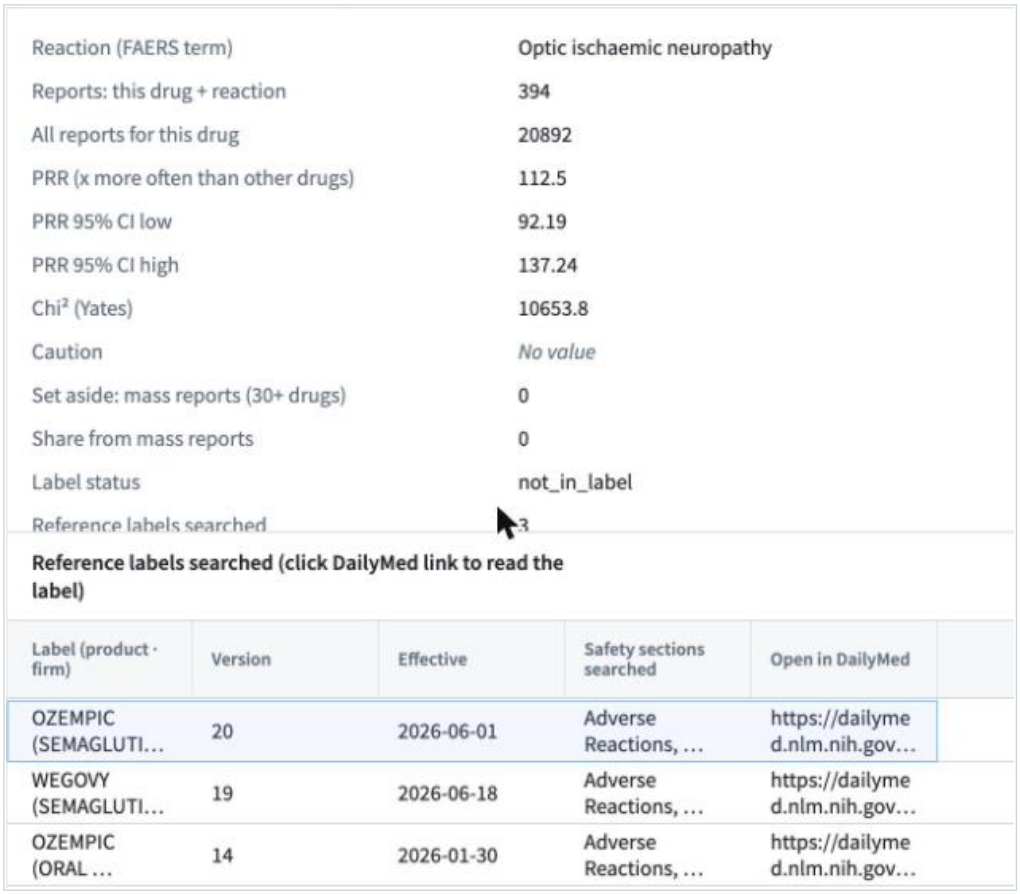


Figure 3. The same signal on page 1 of the app (right panel, cropped): 394 reports, PRR 112.5, label status not_in_label, and the panel *Reference labels searched* with the 3 labels and their DailyMed links.
Source: docs/11-handin/screens/app-1c-labels-searched-naion.jpg (Foundry, 2026-11-05)

How we know

PRR equals the October prototype on the same quarters (A012).

Baseline gate: 5 random ranked signals re-derived by hand; `a` recounted from FAERS rows matched 5/5, and the term was in no reference-label safety section (`baseline-gate.md`).

09

What runs incrementally, and what it saved

Figure 4 (sideways, after chapter 10) draws both pieces: full vs incremental FAERS, and DailyMed XML reuse.

#	Pipeline	Before	After	Helped?
O1	AI extraction cache	9 model calls, 70,776 tokens	0 calls, 0 tokens on rebuild	Yes
O2	Incremental FAERS (A015, D044)	Full rebuild 32 s	Adding Q2: 62 s state + 26 s outputs = 88 s	**No**
O3	DailyMed XML reuse (D047, proposed)	1,321 downloads, 203 s	2 downloads (1,319 reused), 83 s	Yes
O3a	FAERS column pruning	local 12.3-13.0 s	local 12.2 s	**No**

Incremental FAERS (`clean_faers_incremental.py`) keeps a *state* (current cases, suspects, reactions) and a per-quarter log. Each build adds the oldest new quarter with the D022 rule; if drug identity changes, it replays every quarter as a new generation. It shares `suspects()` and `finish()` with the full rebuild, so the two paths cannot drift. Its outputs are identical to the full rebuild (EXCEPT ALL both ways = 0 rows), but at two quarters it is slower: reading back and rewriting the state costs more than parsing one quarter saves. So **signals still read the full rebuild** OBSERVED (O2). Re-measure at 4+ quarters.

DailyMed reuse reads its own previous output and downloads only new versions and non-XML replies (the 2 DQ42 labels, re-tried on purpose). The index is still pulled every build, so a new version is fetched the same day. The baseline came from an earlier build, not back to back (O3).

FAERS column pruning (O3a) read fewer columns from the FAERS files. Locally it did not change the run time, so it was reverted.

> **How we know**
> Build times from the Foundry Builds page; row equality from SQL; see `docs/08-optimization-log/optimization-log.md`. The two *No* rows stay in the log on purpose.

10

Change feed and monitoring

PROPOSED	D045 (change feed) and D046 (monitoring) are built and tested but still await the project owner's OK (the person accountable for this build). Until then they are working pilot features.

Figure 5 (sideways, after Figure 4) shows the two ways a change is found.

`ontology_changes.py` writes one row per change to a signal, recall or label. **Source-dated** events give real history from the first build: 900 recalls reported, 767 terminated, 1,231 label versions effective (the 1,231 of 1,321 label versions that carry an effective date) = 2,898 events. **Snapshot-diff** events compare watched fields (label status, ranked, bucket, caution; recall status and class; label version) with the previous build. A rebuild with the same data added 0 events, as it should OBSERVED. *Since her last review* is a date filter on page 3, not a stored bookmark DESIGNED. The Morning brief card counts the last 30 days (26 events).

`monitoring.py` appends 21 checks per build to `monitoring_checks` (keys, row-count drops over 20%, freshness per source, AI grounded rate ≥ 0.95, cache hits, FAERS reporter-type blanks, ranked signals with no reference label) and rebuilds `monitoring_drift` as a **historical replay** from source dates (last 4 weeks vs the 4 before). Foundry Data Health adds 4 checks. Two real warnings so far: reporter type blank 34% → 72%, and recalls 10 vs 3.

How we know

Locally, a planted change (one signal edited, one recall removed) produced exactly 3 events (A016). Monitoring has run 2 times in Foundry (A017).

Freshness at the gate: openFDA 8 days, FAERS 130 days. For openFDA that is normal for a weekly feed. For FAERS it is the normal lag of a quarterly file, and it means the signal list is about four months old on any given morning.

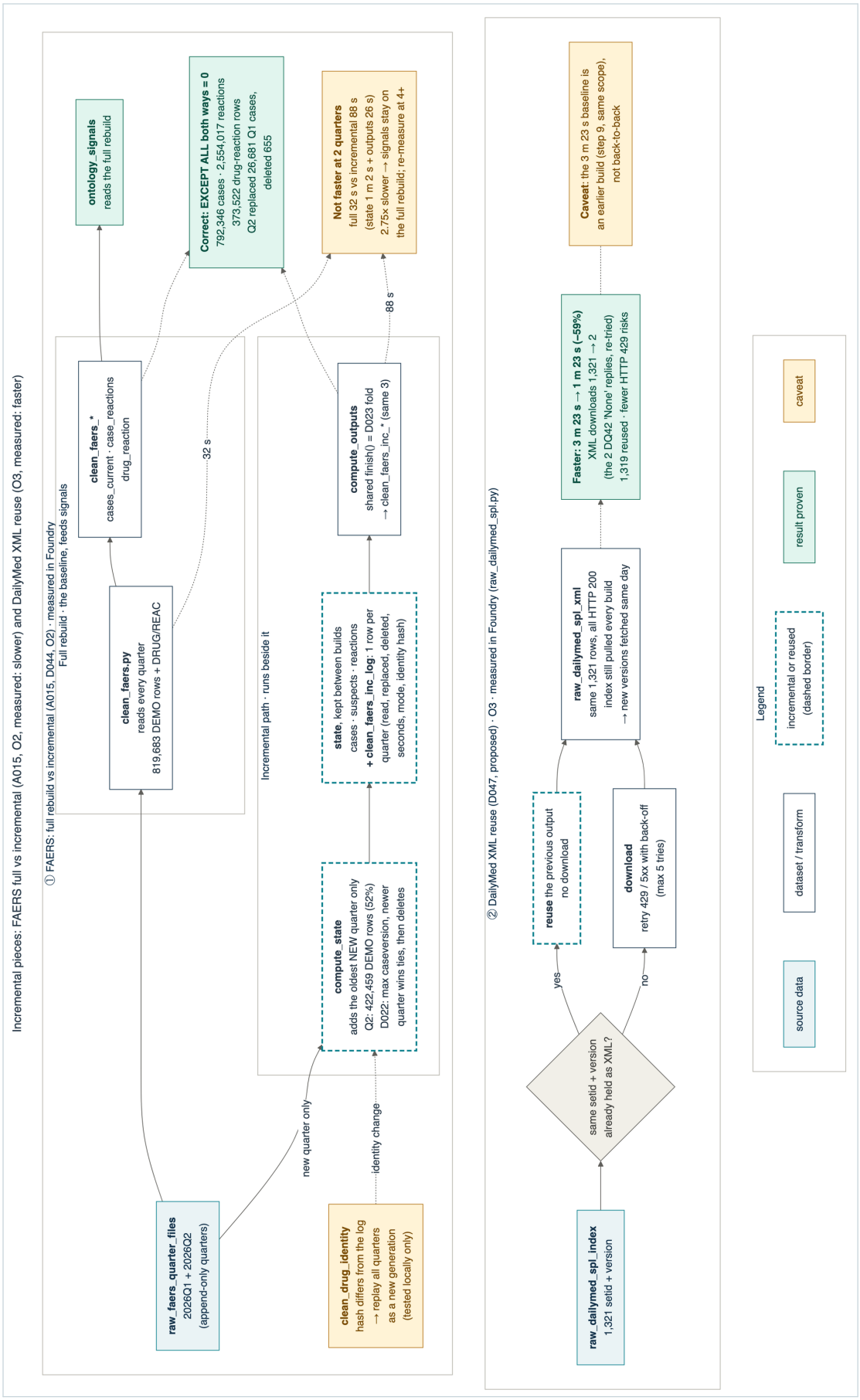


Figure 4. Full vs incremental FAERS (top) and DailyMed XML reuse (bottom), v2 with O3 measured. *DEMO* is the FAERS case file (one row per case version, with the patient's age and sex). (Printed sideways on its own page so the labels can be read.)

Source: docs/diagrams/png/33-handin-incremental-faers-v2.png (source docs/diagrams/src/33-handin-incremental-faers-v2.mmd)

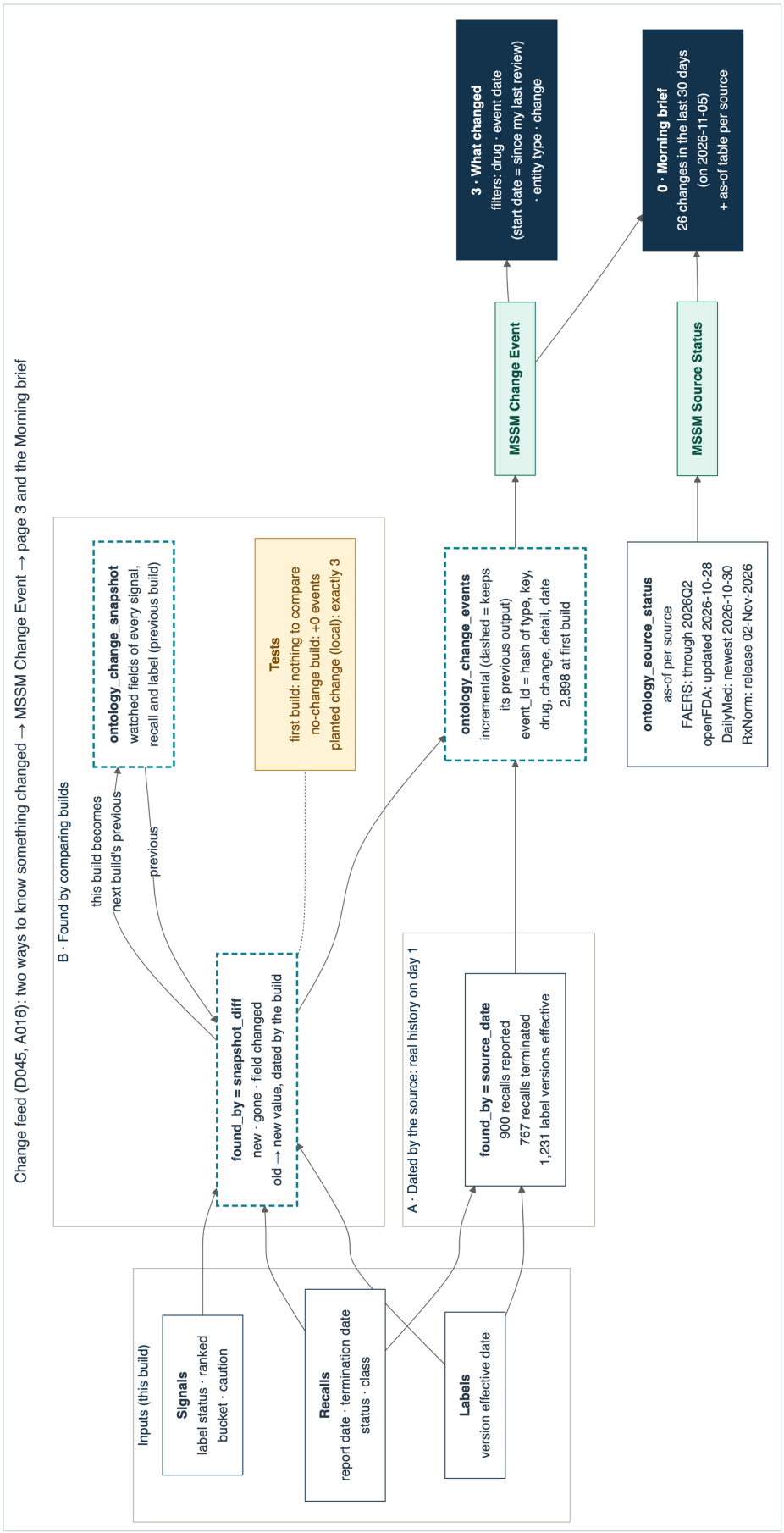


Figure 5. Two ways to know something changed: dates the source carries (A) and comparing this build with the last (B). Both feed MSSM Change Event, page 3 and the Morning brief. (Printed sideways on its own page so the labels can be read.)

Source: docs/diagrams/png/34-handin-change-feed-v1.png

11

How another engineer extends this

Every rule lives in one file. Change it in the GitHub mirror (`foundry/`), test with the local runner, paste into Foundry, commit, then build **in dependency order**. The ripple map (Figure 6, the sideways page after the table) shows what moves.

Change	Edit this	Rebuild, in order	What else moves
Add a drug	`INGREDIENTS` in the 4 API pull files (`raw_openfda_enforcement.py`, `raw_openfda_products.py`, `raw_dailymed_spl.py`, `raw_rxnorm.py`) and `FORMULARY` (RxCUI, name, class) in `clean_drug_identity.py`	raw pulls → identity → labels, FAERS, recalls → `ontology_objects` → signals → changes → monitoring	Check raw totals per drug. If the drug has no reference label, nothing is shown as *not in label* (G4)
Add a source	New Data Connection source + egress policy; a raw transform in `pipelines_raw_ingest`; a clean transform in `pipelines_clean` that joins on `ingredient_rxcui` or `product_key`	new raw → new clean → the curated or signal step that uses it	Write its contract (grain, key, refresh) first; add a freshness check in `monitoring.py`
Change a synonym	A row in `label_synonyms.py`; bump `SYN_VERSION`	`ontology_signals` → changes → monitoring	Label check string changes → every `evaluation_id` changes; old decisions keep the old id. The past-decisions table shows that id, so a mismatch is visible, but the screen does not flag it yet **PLANNED**
Change a bucket	The regex in `BUCKETS` in `ontology_signals.py`	`ontology_signals` → changes	Only the ranked queue moves; evaluation ids stay, so decisions stay linked
Change a threshold	`MAX_DRUG_ROWS` or the Evans test in `ontology_signals.py`; bump `RULE`	`ontology_signals` → changes	New rule → new `signal_key`s; record a new decision version
Change identity	A salt word or name rule in `clean_drug_identity.py`	identity → FAERS (full; the incremental state replays) → signals	Suspect rows change for every quarter

Three habits that save an afternoon

Data Lineage *build all ancestors* re-runs every raw pull and the 700 MB FAERS download. Build file by file.

The repository Build button may skip an unchanged incremental transform: build from the dataset page (Build ▾ → Force) and check History.

Record a changed rule as a new decision version in DECISIONS.md; never edit one in place.

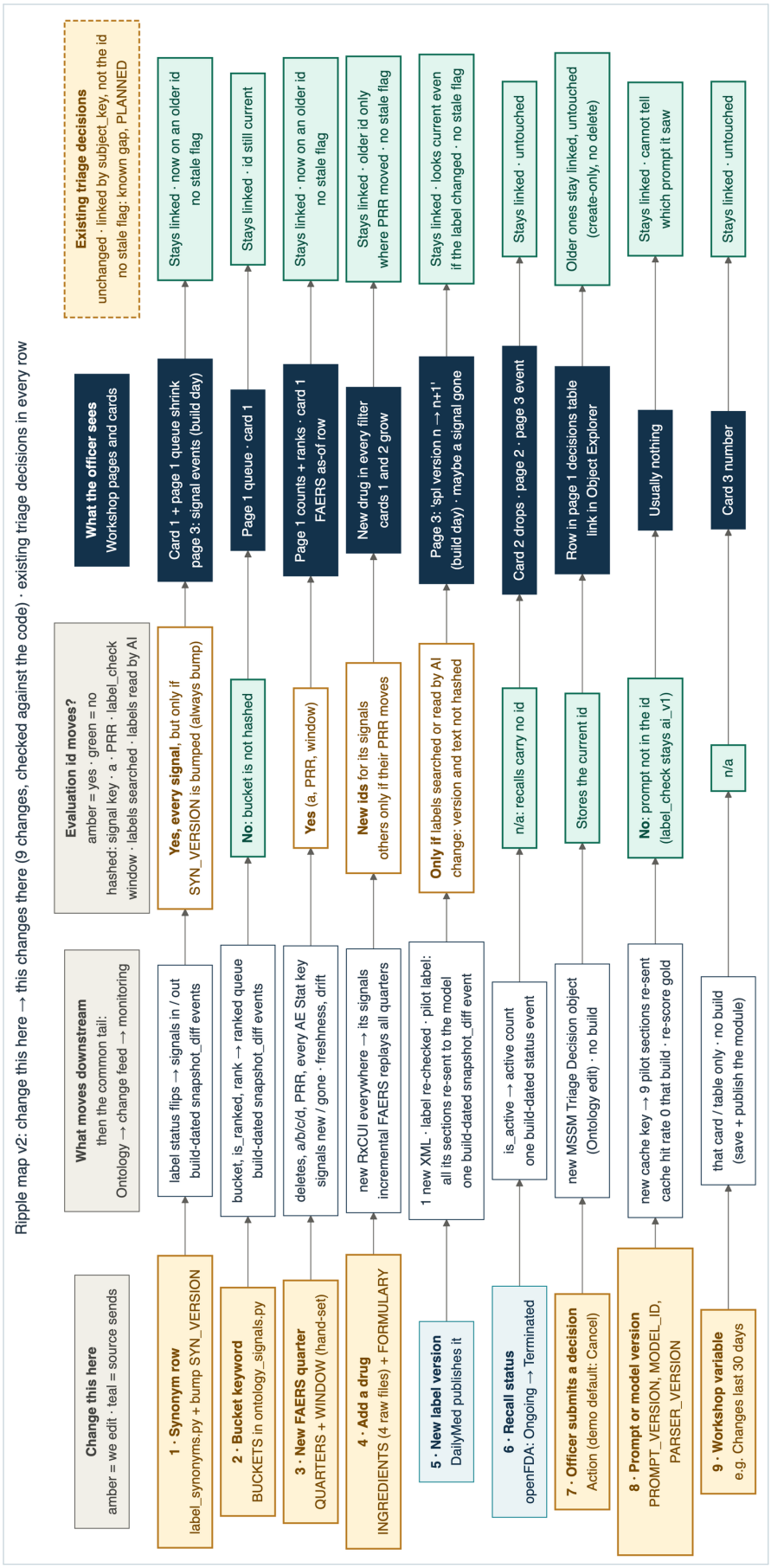


Figure 6. Ripple map v2: nine changes, each traced to what moves downstream, whether the evaluation id moves, what the officer sees, and existing triage decisions (they stay linked in every row; there is no stale flag yet, a known gap). Checked against the code (docs/11-handin/demo/ripple-table.md). (Printed sideways on its own page so the labels can be read.)

Source: docs/diagrams/png/37-handin-ripple-v2.png

Gate: Chapter 11: another engineer can extend it without asking TESTED

Required evidence: Each common change names the file to edit, the rebuild order and its side effects.

Result: Met for six changes. The ranked queue was 1,355 at step 9; the first synonym table took it to 1,210; two later changes were done for real with this guide: buckets D016 v3 (1,210 → 1,115) and synonyms syn_v2 (1,115 → 1,073) OBSERVED.

If it fails: Add the missing step to `foundry/README.md` (How a change reaches Foundry).

A

Where the numbers come from

Figures quoted from project records (not in numbers.json)

These come from earlier measured records, not from the baseline-gate capture. They are quoted with their source.

Figure	Source
openFDA OR search 11 vs 25 recalls; `openfda.*` 47 vs 141 applications	DECISIONS learnings, 2026-11-04
Morphine substring match about +50% rows	DQ34, data-issues register (prototype, 2026-10-20)
Oxycodone 4,464 → 3,408 cases after fold; 465 mass reports, median 122 drugs	foundry/README.md; D023 v2; A012
Primary-suspect-only would miss 56% of oxycodone cases	DECISIONS learnings, 2026-11-04
NAION: 36 semaglutide safety sections, none mention it	A014; DECISIONS learnings, 2026-11-05
20 of 31 printed oxycodone recall NDCs not in the directory	DECISIONS learnings, 2026-11-04
Planted change → exactly 3 events	A016
Oxycodone 4,464 → 3,408 equals the October prototype	DECISIONS learnings, 2026-11-04
2,533 stats meet Evans; 1,427 / 8 / 163 stats in label by text / AI / synonym; 211 stats possibly; 231 signals with the mass-report caution	Diagram 32 v2 (docs/diagrams/src/32-handin-signal-pipeline-v2.mmd)
Ozempic v20, Wegovy v19, oral Ozempic v14 (NAION reference labels)	screens/README.md (app-1c); diagram 32 v2

Numbers used in this report

Every number in the table below is looked up from `docs/11-handin/numbers.json` at build time. Source: Foundry SQL (repo SQL panel) after the baseline-gate rebuild of the whole downstream chain: raw_dailymed_spl → clean_labels → clean_label_product_map → ontology_objects → ai_label_extraction → ontology_signals → ontology_changes → monitoring. Build durations from the Foundry Builds page. Query: docs/11-handin/baseline-gate.md. Captured: 2026-11-05T07:50Z. Figures quoted from other project records (decision log, data-issues register, optimization log) name that record where they appear.

Key	Value
`ai.d015_target_f1`	0.85
`ai.gold_f1`	0.97

Key	Value
ai.gold_recall_strict	35/35
ai.grounded	448
ai.model_calls_on_rebuild	0
ai.pilot_labels	2
ai.reactions	448
ai.sections	9
app.labels_searched_naion	3
changes.label_version_effective	1,231
changes.last_30_days	26
changes.recall_reported	900
changes.recall_terminated	767
changes.snapshot_diff_after_rebuild	0
changes.source_date	2,898
changes.total	2,898
faers.case_reactions	2,554,017
faers.cases_current	792,346
faers.drug_reaction_rows	373,522
faers.q2_cases_deleted	655
faers.q2_cases_replaced	26,681
labels.total	1,321
monitoring.checks_per_build	21
monitoring.data_health_checks	4
monitoring.faers_age_days	130
monitoring.openfda_age_days	8
monitoring.recalls_last4w	10
monitoring.recalls_prev4w	3
monitoring.reporter_type_blank_q1	0.343
monitoring.reporter_type_blank_q2	0.721
monitoring.runs	2
optimization.O1.after_calls	0
optimization.O1.after_tokens	0

Key	Value
optimization.O1.before_calls	9
optimization.O1.before_tokens	70,776
optimization.O2.full_rebuild_s	32
optimization.O2.incremental_add_q2_s	88
optimization.O2.outputs_s	26
optimization.O2.state_s	62
optimization.O3.after_downloads	2
optimization.O3.after_s	83
optimization.O3.before_downloads	1,321
optimization.O3.before_s	203
optimization.O3.reused	1,319
optimization.O3a.local_run_after_s	12.2
optimization.O3a.local_run_before_s	12.3-13.0
raw.dailymed_labels	1,321
raw.drugsfda_rows	663
raw.ndc_rows	1,824
raw.recall_rows	930
raw.recalls	900
recalls.active	126
recalls.newest	D-0040-2027
recalls.newest_reported	2026-10-21
recalls.total	900
scope.drugs_formulary	13
scope.glp1	5
scope.opioids	8
signals.ae_stats	14,921
signals.naion.a	394
signals.naion.all_reports_drug	20,892
signals.naion.chi2_yates	10653.8
signals.naion.prr	112.5
signals.naion.prr_ci_high	137.24

Key	Value
signals.naion.prr_ci_low	92.19
signals.naion.rank	1
signals.naion.reaction	Optic ischaemic neuropathy
signals.not_in_label	1,844
signals.possibly_in_label	108
signals.ranked	1,073
signals.ranked_history.d016_v3	1,115
signals.ranked_history.step9	1,355
signals.ranked_history.syn_v1	1,210
signals.ranked_history.syn_v2	1,073
signals.total	1,952
synonyms.high	69
synonyms.medium	72
synonyms.rejected	117
synonyms.rows	258
synonyms.version	syn_v2